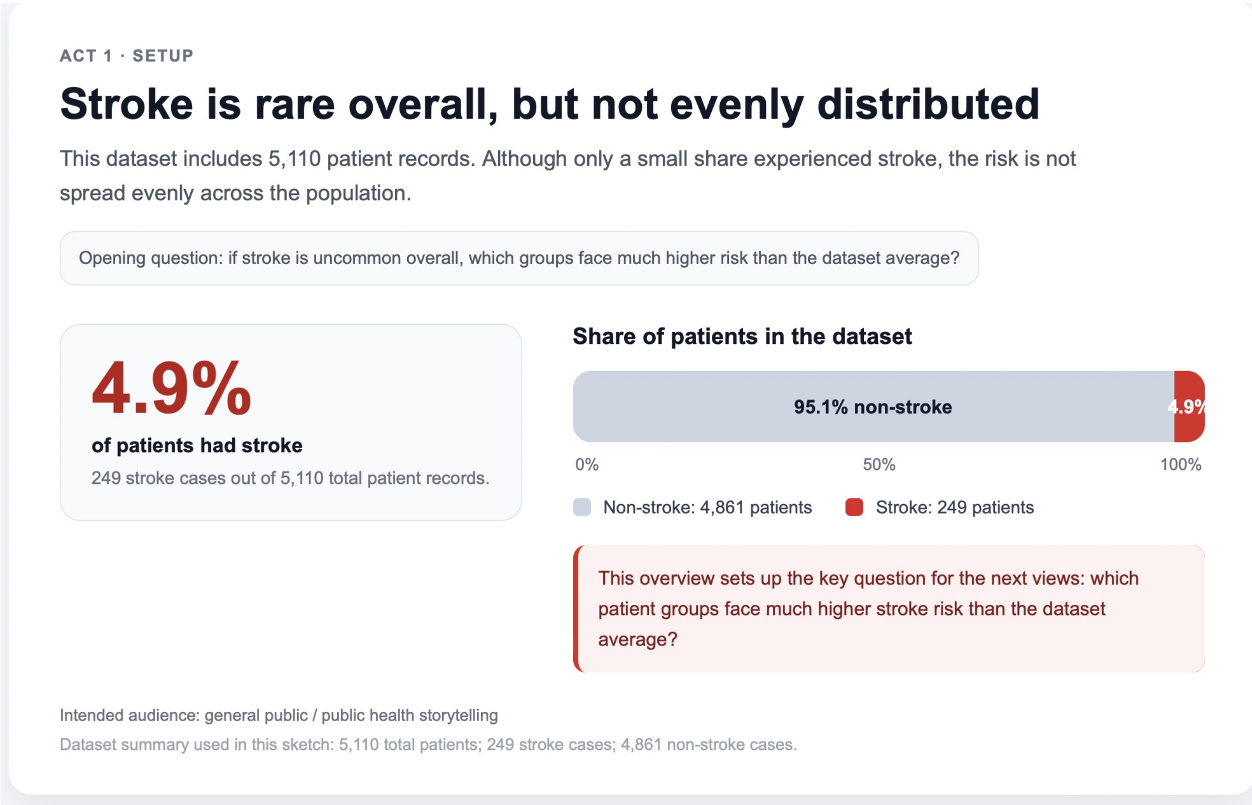
Storytelling Visualization Redesign of the Healthcare Stroke Dataset

This visual story is designed for a general audience. It begins with background context, then moves into the clearest evidence about stroke risk, and ends with a concise summary of the highest-risk groups. The sequence follows a three-act structure: setup, detailed views, and conclusion.

Dataset used: healthcare stroke dataset (5,110 patient records).

Chart 1. Stroke is rare overall, but not evenly distributed



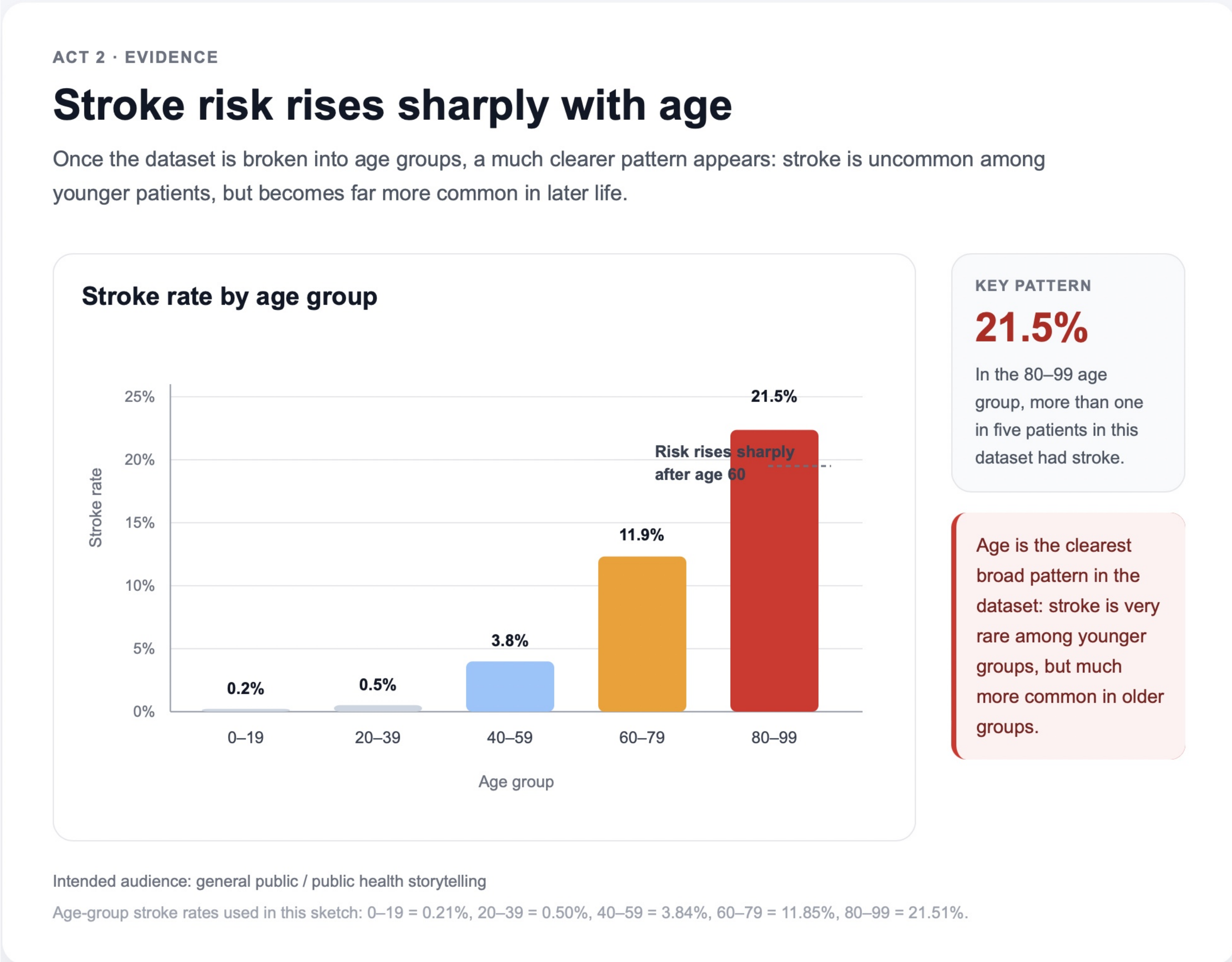
(a) What this chart is intended to show the audience.

This chart introduces the dataset by showing that stroke cases are relatively rare overall compared with non-stroke cases. It serves as the set-up/background of the story by establishing the overall context before asking which groups face much higher risk than the dataset average.

(b) What I changed from the AI's original suggestion, and why.

I changed the original idea from a simple two-bar comparison to a headline number with a 100% stacked bar so that the first view would function more clearly as the setup of the story rather than just a descriptive chart. This revision makes the class imbalance easier for a general audience to grasp immediately and also frames the central narrative question for the later views: if stroke is uncommon overall, which groups account for the higher-risk cases?

Chart 2. Stroke risk rises sharply with age



(a) What this chart is intended to show the audience.

This chart shows that stroke risk increases sharply across older age groups, making age the clearest overall pattern in the dataset. It provides the first major piece of evidence in the story by showing that stroke is very uncommon among younger patients but becomes far more common in later life.

(b) What I changed from the AI's original suggestion, and why.

I kept the AI's suggestion of using age as a central view, but I refined it into an annotated bar chart because age groups are discrete categories and a bar chart makes the increase in stroke risk easier to compare than a more complex layout. I also added annotation so that this chart would not only display the pattern, but clearly communicate its narrative purpose as the first major piece of evidence in the story.

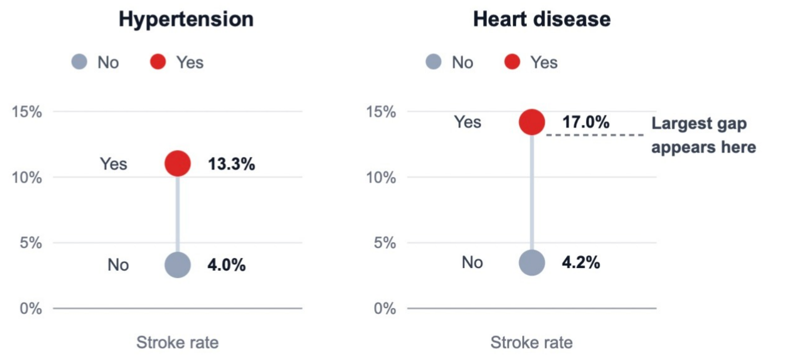
Chart 3. Cardiovascular conditions further increase risk

ACT 2 · EVIDENCE

Cardiovascular conditions further increase risk

Age is the broadest pattern in the dataset, but it is not the whole story. Patients with hypertension or heart disease show substantially higher stroke rates than those without these conditions.

Stroke rate by health condition



Patients with hypertension show a visibly higher stroke rate. The gap is even larger for patients with heart diseas

KEY PATTERN

17.0%

Patients with heart disease had the highest stroke rate among the groups shown in this view.

After age, the clearest additional pattern is that stroke is more common among patients who already have hypertension or heart disease.

Intended audience: general public / public health storytelling

Stroke rates used in this sketch: hypertension = no 3.97%, yes 13.25%; heart disease = no 4.18%, yes 17.03%.

(a) What this chart is intended to show the audience.

This chart shows that hypertension and heart disease are both associated with substantially higher stroke rates than their corresponding no-condition groups. It deepens the story by showing that age is not the only important pattern: existing cardiovascular conditions further increase risk.

(b) What I changed from the AI's original suggestion, and why.

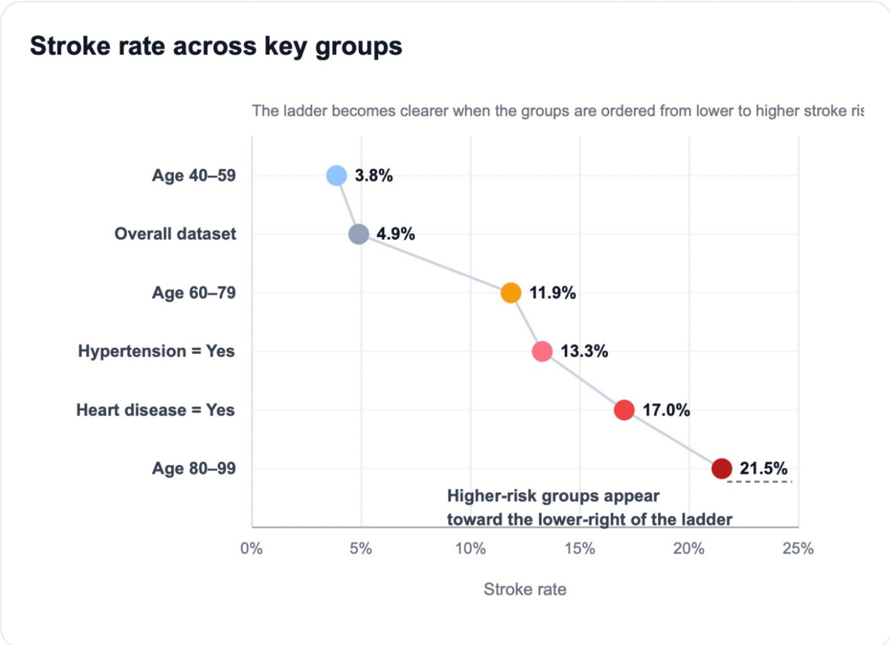
I changed the original grouped-bar idea into two small paired-dot plots because this format makes the yes-versus-no gaps more visually immediate while reducing the clutter of a second bar-chart view. This change also helped the middle section of the story progress more effectively: instead of repeating the age chart's visual form, it introduces a cleaner comparison that deepens the narrative from broad age patterns to condition-specific risk.

Chart 4. A clear high-risk ladder emerges

ACT 3 · CONCLUSION

A clear high-risk ladder emerges

Taken together, the earlier patterns point to a clear conclusion: stroke is not evenly distributed in the dataset. Risk is much higher among older patients and among patients with cardiovascular conditions.



MAIN TAKEAWAY

21.5%

The oldest age group in the dataset shows the highest stroke rate, followed by patients with heart disease and hypertension.

The final takeaway is not that everyone faces the same stroke risk. Instead, the risk is concentrated among clearly identifiable higher-risk groups.

Intended audience: general public / public health storytelling

Group rates used in this sketch: age 40–59 = 3.84%, overall = 4.87%, age 60–79 = 11.85%, hypertension = yes 13.25%, heart disease = yes 17.03%, age 80–99 = 21.51%.

(a) What this chart is intended to show the audience.

This final chart summarizes the clearest high-risk groups in the dataset and shows that stroke is concentrated among older patients and patients with cardiovascular conditions. It functions as the conclusion of the story by bringing together the earlier findings into a single summary view.

(b) What I changed from the AI's original suggestion, and why.

I changed the original text-only ending into a ranked summary chart because I wanted the conclusion to remain visual and evidence-based rather than shifting abruptly into prose. I also reordered the groups from lower to higher risk so that the final view would read more clearly as a summary ladder, allowing the audience to leave with a single, memorable takeaway about which groups face the highest stroke risk.